

Problem 1:

Imagine you are an editor at a large publishing company. They need to change the ending of a story to make it a palindrome. The ending is a string of lowercase letters, but it isn't a palindrome yet. To make it perfect, the editors want to change the string with as few changes as possible and make sure the result is the **lexicographically smallest palindrome**.

You are responsible for helping them turn the ending into a palindrome with the **fewest** changes. If there are multiple ways to do this, you should choose the one that is lexicographically smallest.

(Tips: For two strings a and b of the same length, if they differ at the first position where they are not the same, and the character in a at that position appears earlier in the alphabet than the character in b, then we consider the dictionary order of a to be smaller than that of b.)

Input:

A non-empty string

Output:

Return the final palindrome string with the fewest modifications. If there are multiple solutions, return the lexicographically smallest one.

Example 1

Input:

"sunday"

Output:

"saddas"

Explanation:

"saddas" is a palindrome string, with the fewest modifications and the lexicographically smallest order.

Example 2

Input:

"seven"

Output:

"neven"

Explanation:

"neven" is a palindrome string, with the fewest modifications and the lexicographically smallest order.

Problem 2 — Mao's Lucky String

A type of string is called a "Mao's Lucky String", defined as follows:

- The index starts from 0.
- Characters at even indices (0, 2, 4, ...) must be vowels (a, e, i, o, u).
- Characters at odd indices (1, 3, 5, ...) must be consonants (the remaining 21 lowercase English letters).

Example: abi is a lucky string (a is at position 0, b is at position 1, i is at position 2).

Repeated characters are allowed, and the string may start with any valid letter.

Now, Mao has locked his chocolate ice cream in the refrigerator.

The lock displays a number n ($1 \leq n \leq 10$).

The password is the number of possible “Mao’s Lucky Strings” of length n .

Please help Mao compute this number.

Input :

A single integer n .

Output Format

A single integer — the number of possible lucky strings.

Example

Input

1

Output

5

Explanation

When $n = 1$, only vowels can appear, so there are 5 possibilities: a, e, i, o, u.

Problem 3: Gold Coin Pact

Mao has a chocolate bar of length n . He wants to break it into several smaller pieces of **different lengths**, such that the **product of all lengths** is as large as possible — the larger the product, the more he earns!

Can you help Mao find the most profitable way to break the chocolate bar?

Input

A single integer n ($3 \leq n \leq 10000$).

Output

Output one possible decomposition of n into different positive integers whose product is maximized.

The numbers should be separated by a single space and listed in ascending order.

Example Input

10

Example Output

2 3 5

Problem 4: The Mirrored Code

Mao discovered a magical “number mirror.” When he enters a digit string, the mirror treats it as a number in base p , converts that number into base m , and then **reverses the order of the base- m digits** before

displaying. Leading zeros in the reversed result should be dropped (i.e., output starts from the highest non-zero digit in the reversed string).

Input

One line containing integers p , m and a string k , where:

- $2 \leq p, m \leq 20$
- k is a valid base- p number (digits use 0-9 and A-J for 10-19; uppercase)

Output

Output the mirrored result: convert k from base p to base m , then reverse that base- m string and remove any leading zeros from the reversed string. Use A-J for digits 10-19.

Example:

Input:

8 12 157

Output:

39

Explanation:

$157_8 = 64 + 40 + 7 = 111$ (decimal).

111 in base 12 is "93". Reverse it $\rightarrow$ "39". Output **39**.

Problem 5: Puzzle Game

As 1024 approaches, Kuan gives Mao a carefully prepared gift, locked inside a box.

On the lock are two extremely large numbers, with a note saying: "Multiply them to get the password."

Mao exclaims: "These numbers are longer than my hair!"

Please help Mao compute the product of these two huge numbers to unlock the box.

Input

Two lines, each containing a non-negative integer.

Output

A single non-negative integer representing their product.

Example Input

23

456

Example Output

10488

Hint

Each number can be as large as 10^{2000} , so standard integer types (like long long) are not sufficient.